

Nakhon Si Thammarat City

...Voluntary Local Review



Dr. Kanop Ketchart
Mayor of Nakhon Si Thammarat City
March 31, 2023



Outline

**01
City
Overview**

**02
Effective
VLR
Conduct**

**03
Challenges &
Opportunities**

**04
Sustainable
Development**

**05
SDG Projects**

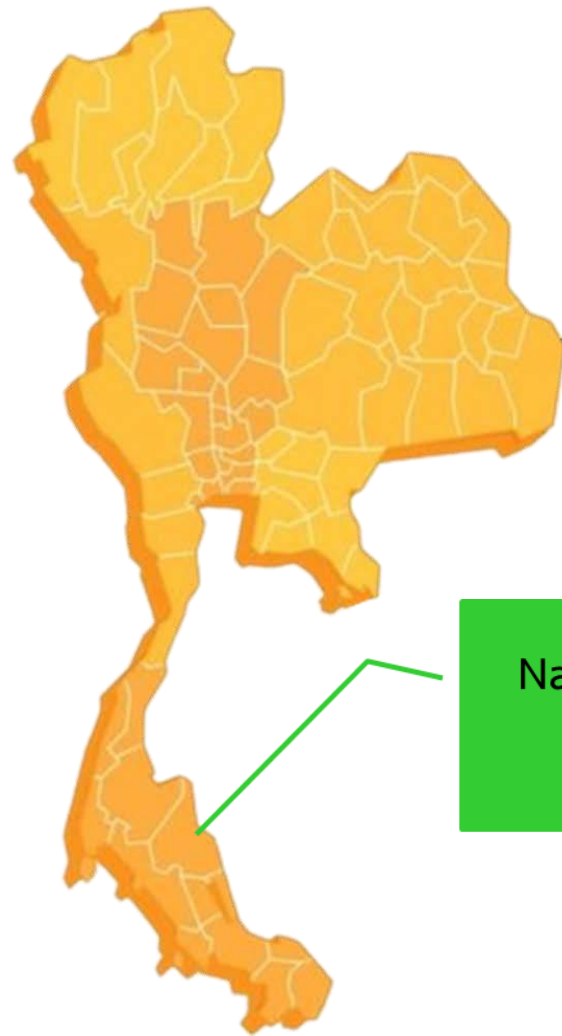




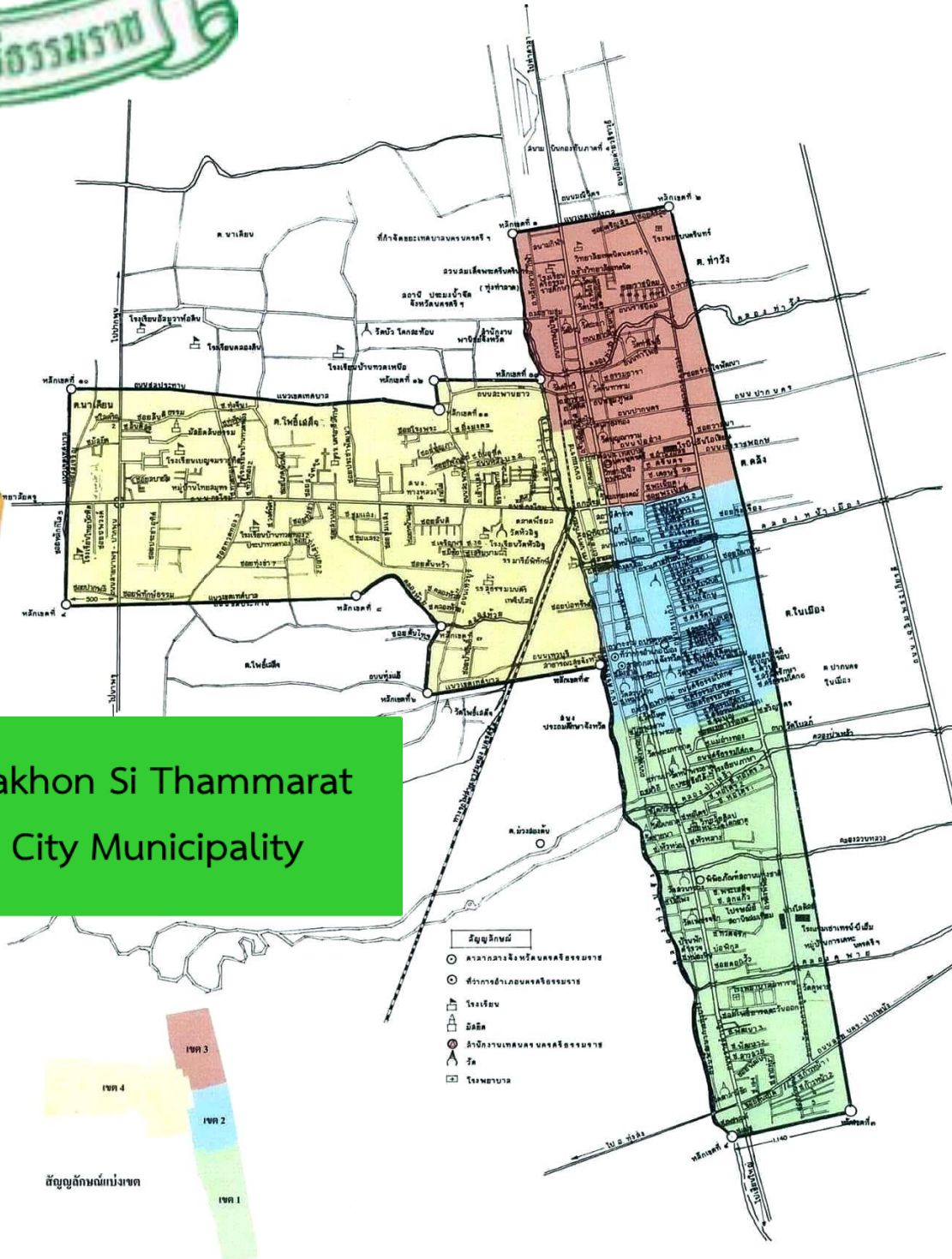
Nakhon Si Thammarat City Municipality

Nakhon Si Thammarat province, Thailand

- ✓ Located in the south of Thailand
- ✓ More than 1500 years old city, “Ligor City”
- ✓ Area of 22.56 km² / population > 100,000 people
- ✓ Center of education (30,000 students) and cultural tourism (4 million visitors per year)
- ✓ Development with a balanced-combination of prestigious history and remarkable urbanization based on a people-centric strategy



Nakhon Si Thammarat
City Municipality





Effective VLR Conduct

How?

- orderly/methodically data collection with proper technology applied
- farsighted mayor and executives
- well-cooperated officers
- suitable announcement/acknowledgement
- public participation which includes all groups of residents

Voluntary Local Review 2022

The Implementation of the
UN Sustainable Development Goals
in Nakhon Si Thammarat City Municipality



Challenges & Opportunities with VLR implementation

Challenges

- brand new task : no experience

Opportunities

- equality for all : ensuring infrastructure and public services are accessible by ALL
“Leave No one Behind”
- provide well-organized information for tourists and investors



Sustainable Development

- VLRs help cities/local governments realize their overall work/progress (mistakes, strength & weakness and movements towards sustainable development)
- VLRs enhance better cooperation/connection among officers
- VLRs broaden acknowledgement as well as stimulate public participation



SDG Projects
City of Nakhon Si Thammarat

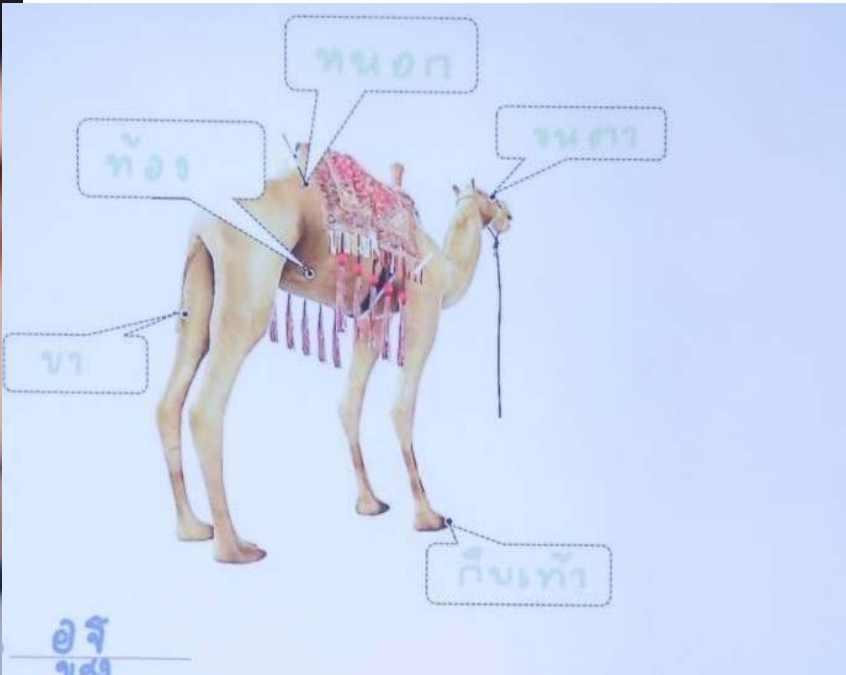




SDG 4: Metaverse Classrooms

(Quality Education)

- **10 Municipality-Affiliated Schools**
- **Trained 300 teachers to use AR and VR in schools with 11,000 students**





SDG 4: Mayor Classroom

(Quality Education)

- School for underprivileged students
- Free transportation, meals, special classroom
- Enhancing “Education for All”





SDG 9: City E-Service

(Industry, Innovation and Infrastructure)

- Winner of “Thailand Smart City Solutions Awards 2022”
- Residents send requests to the City via two-way LINE digital platform
- Solved each request within 48 hours
- 10,385 services completed (99.11 %)
- 42,616 members (40% of city residents)

as of Mar 29, 2023



SDG 9:City E-Service

(Industry, Innovation and Infrastructure)



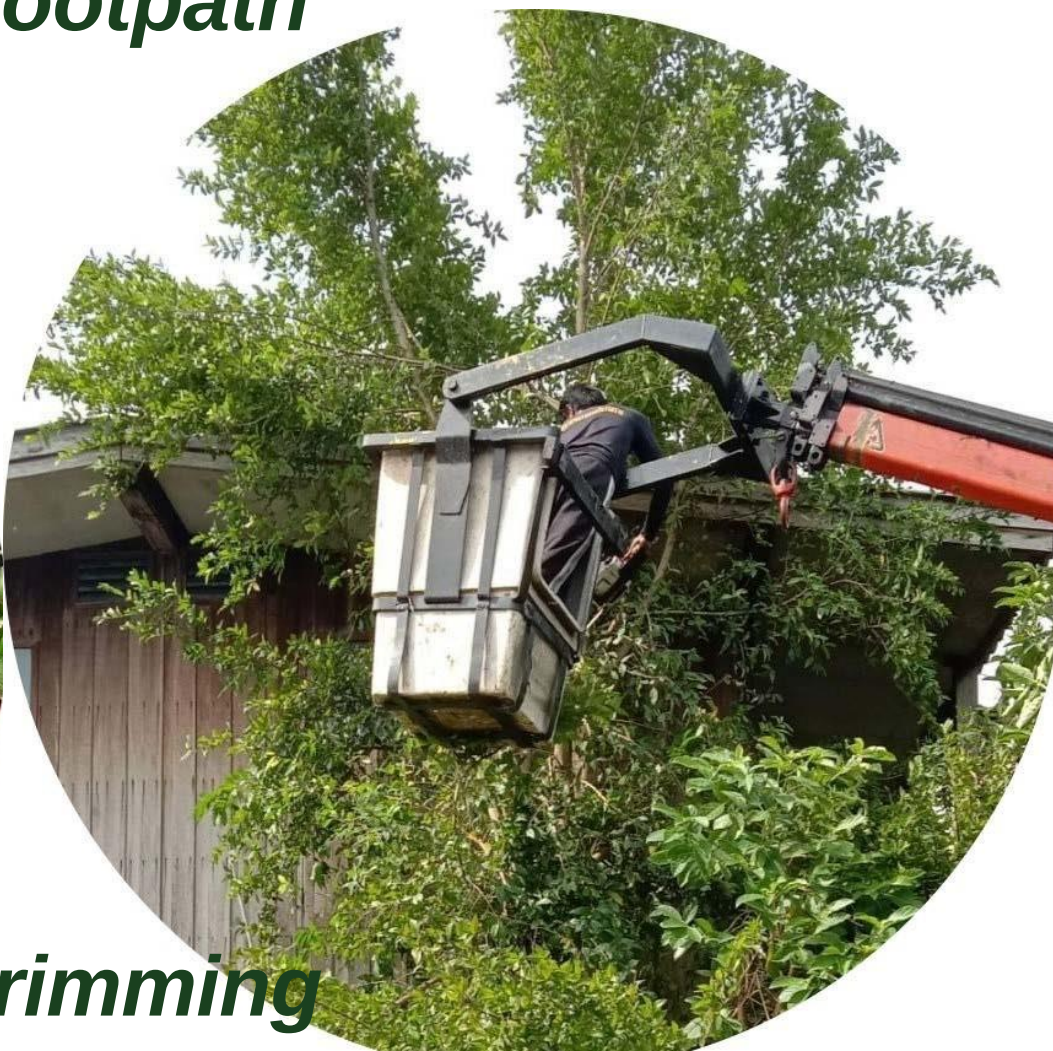
Street Avenue and Footpath



Public City Light



Tree Trimming



Public Plumbing



SDG 9: Flood Early Warning System

(Industry Innovation and Infrastructure)



Asian Insider

Thai city removes bins to reduce pollution



Tan Hui Yee
Indochina Bureau Chief

NAKHON SI THAMMARAT – In the southern Thai city of Nakhon Si Thammarat, trash bins used to “walk”.

The municipal authority left waist-high bins on the street for households to deposit their bagged trash. But residents, trying to rid themselves of this eyesore, would push the wheelie bins away from their homes – to the front of their neighbours’ homes – sparking furious rows.

Bounced from place to place, the bins would eventually end up in front of vacant lots, attracting vermin as well as passers-by who dumped loose bits of litter.

“Nobody wanted bins in front of their house. And they argued a lot,” said Ms Kamolnun Taninthong, the leader of the 1,000-strong Samrit Prasong community in the city. “And they kept pushing it here and there, and pushing it outside (the community), where strangers would throw their trash.”

Such dumping practices increased the incidence of stray litter and worsened plastic pollution that threatened the marine environment in the Gulf of Thailand because the city had a network of canals, and sat next to the coast.

In 2020, Nakhon Si Thammarat was one among four cities in South-east Asia that were chosen for a project by the United Nations Economic and Social Commission for Asia and the Pacific (ESCAP) and the Japanese government, to tackle plastic waste pollution in rivers and oceans. The other three cities were Da Nang in Vietnam, Kuala Lumpur in Malaysia and Surabaya in Indonesia.

Nakhon Si Thammarat then had the largest accumulation of unmanaged solid waste in Thailand. It generated 9,000 tonnes of plastic waste a year, with nearly half of its mass made up of plastic bags. Every year, some 90 tonnes of its plastic waste entered its waterways, while another 362 tonnes were left on land or in drains. Left unmanaged, they broke down into microplastics that further contaminated the environment.

Part of the city’s problem was that waterways that connected the highlands of the surrounding province to the sea passed through many districts before reaching the municipal area, accumulating trash along the way.

This is a common problem in South-east Asia, where the concentration of microplastics in seawater can range very widely from 0.1 to 11,100 pieces per litre, according to a 2021 review article by Dr Emily Curren from the National University of Singapore.

Locations with greater human

activity have a higher abundance of microplastics.

“Regions with more lax disposal of waste tend to see higher abundance of microplastics in their coastal areas. The government and the authorities have to intercept this at the source and enforce proper disposal of waste,” Dr Curren told The Straits Times. This applies not only to households but to businesses. Illegal disposal of wastewater by a textile factory, for example, could cause a spike in the amount of textile microfibres, which are essentially microplastics, in the water.

Working with environmental experts, Nakhon Si Thammarat attempted a solution that seemed counterintuitive – removing the wheelie bins. In Samrit Prasong, residents placed their bagged trash on the street directly for collection, at specific times agreed with municipal workers.

Ratchadamnoen Road, a large road spanning 10km of Nakhon Si



Under Nakhon Si Thammarat’s programme to reduce trash leakage, workers remove canal trash daily. The amount of waste scooped up from canals has fallen by half since the city started this activity in October 2022. ST PHOTOS: TAN HUI YEE

Thammarat city, was stripped of 90 per cent of its 400 wheelie bins. Trash collection times were staggered according to the convenience of affected business owners.

A large open container normally sited downtown for bulky waste was removed, and replaced with a house-to-house collection service. Meanwhile, 10 schools within the city installed water dispensers to encourage students to stop buying water in disposable bottles.

The result has been a cleaner environment, with less plastic pollution. On a busy Monday morning, this correspondent saw that the wide pavements fronting the 13th

century Phra Mahathat Woramahawihan temple were free of litter, despite the heavy footfall.

The amount of plastic trash accumulated by each school has dropped by 10 to 50 per cent.

The amount of waste scooped up from city canals has dropped by half since the city started cleaning them daily in October 2022.

“We worked together. This project belongs to the people,” Mr Kanop Ketchat, mayor of Nakhon Si Thammarat municipality, told The Straits Times. “Once we removed the bins, all of them loved it. It was better for the economy and tourism, we got positive energy and kept going.”

Ambitious projects to tackle pollution are common, but what made the difference in Nakhon Si Thammarat was how city authorities got the public to take part by earning its trust, observed Dr Jutamas Kaewsuk, an environmental engineering lecturer from Mahidol University.

Before removing the bins in Ratchadamnoen Road, vice-mayor Vongvachira Ovararin held more than 10 meetings with residents and business owners, and started the project only when the households in each block agreed on a date to begin the new trash collection arrangement.

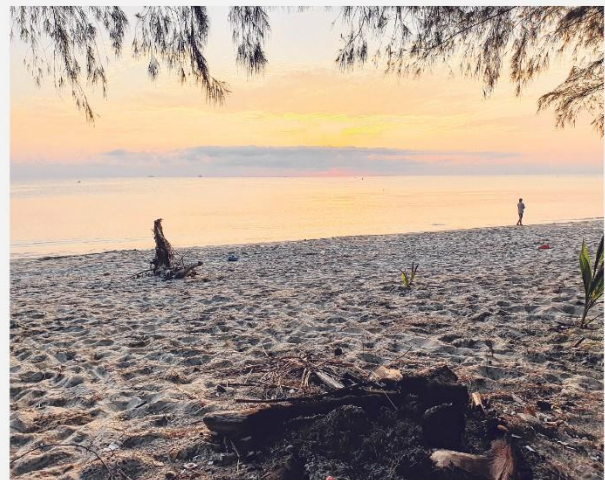
“After each meeting, I walked again down the road to knock on doors to speak to people who did not attend the meeting,” said Dr Vongvachira.

In one community of 100 households, residents asked that the bins that had been removed be put back. Two weeks later – realising that the new arrangement was indeed less of an eyesore and more convenient – they asked the municipality to take them away.

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Garbage trucks turn up at various spots in the city to collect trash at pre-appointed times selected by the communities concerned.



The city’s proximity to the coast means that there is a high chance that its trash, if not properly disposed of, will end up in waterways and the sea.

SDG 11: Clean City

(Sustainable Cities and Communities)

▪ Scheduled trash pickup

▪ Waste sorting at source

▪ Street with no bins

▪ Food waste management

▪ People participation



Thank You

